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Vocalization Type Shapes the Temporal Dynamics of Crossmodal Emotional Word Recognition: An Electroencephalography Study

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ABSTRACT

This study examined how vocalization type, linguistic (spoken words with semantic content) and nonlinguistic (such as laughter or screaming), modulates emotional word recognition and how valence, arousal, and concreteness unfold over time. Using a crossmodal priming paradigm, participants first heard a vocal prime and then categorized a visually presented target word while electroencephalography was recorded. Event-related potential analyses focused on the P1 and early posterior negativity (EPN) components, and time-resolved representational similarity analysis (RSA) was applied to capture distributed neural patterns. Behaviorally, negative words produced longer reaction times compared with neutral and positive words. Incongruent prime–target pairings elicited larger P1 amplitudes, reflecting early mismatch costs. EPN amplitudes were larger for congruent than incongruent pairings, and for negative compared with positive and neutral words. RSA revealed that vocalization type was encoded earliest and briefly around target onset (0–128 ms), arousal emerged earlier (204–568 ms) than valence (512–920 ms), and concreteness showed a sustained representation across stages (188–996 ms). These findings demonstrate that crossmodal vocal priming influences both perceptual and semantic processing and clarifies the temporal organization of arousal, valence, and concreteness in emotional word recognition.

1 | Introduction

Human emotional communication is multimodal, relying on auditory and visual channels to support the perception and understanding of affective meaning (Campbell 2008; Driver and Noesselt 2008). Language is a central medium for emotion, yet vocal cues also carry rich affective information that shapes attention and memory (Frühholz and Grandjean 2013). Such cues include linguistic vocalizations (e.g., spoken words) and nonlinguistic vocalizations (e.g., laughter or sobbing). Through crossmodal enhancement, emotional sounds can influence visual word recognition (Gerdes et al. 2014). However, it remains

unclear how linguistic and nonlinguistic vocalizations differentially shape the neural dynamics of emotional word processing (Doyle and Lindquist 2018).

Vocal communication includes two complementary forms: nonlinguistic and linguistic vocalizations (Brück et al. 2011). Nonlinguistic vocalizations lack lexical content but convey emotional meaning through acoustic features like pitch, loudness, and timbre, directly signaling arousal and valence (Banse and Scherer 1996). Linguistic vocalizations communicate affect through prosodic modulation layered onto symbolic and semantically interpretable speech (Kotz and Paulmann 2007).

In this study, we follow Pell et al. (2015) in defining linguistic vocalizations as speech sounds with identifiable semantic content, excluding pseudo-utterances. This distinction is consistent with the dual-route model of vocal processing, which proposes that affective and linguistic cues are processed in parallel neural pathways (Ethofer et al. 2012).

Although emotional vocalizations can influence lexical processing, the mechanisms by which linguistic and nonlinguistic vocalizations differentially affect semantic activation remain unclear. Nonlinguistic sounds, especially those expressing negative affect, are thought to capture attention through evolutionarily shaped threat detection pathways (LeDoux 2012), which may slow semantic access. Supporting this view, negative musical primes have been found to prolong reaction times in semantic categorization compared with positive primes (Gao et al. 2022). In contrast, emotionally congruent contexts facilitate word processing, with faster and more accurate responses in affective priming tasks using music and sound (Armitage and Eerola 2022; Gerdes et al. 2013; Steinbeis and Koelsch 2011). Importantly, aversive nonlinguistic vocalizations, such as retching sounds, selectively enhance the recognition of negative valence words, showing category-specific priming effects (Scherer and Larsen 2011).

To our knowledge, no previous work has directly compared how linguistic and nonlinguistic vocal primes influence emotional word recognition. Glaser and Glaser (1989) proposed that nonlinguistic sounds can bypass lexical analysis and activate affective–semantic networks more directly, which may produce stronger priming than linguistic cues. Clarifying these differences is important for understanding crossmodal emotional priming. Such evidence could also refine models of audiovisual emotion integration and shed light on how semantic facilitation operates in multisensory contexts. A further step in clarifying these effects is to consider the internal semantic structure of emotional words. These words vary systematically along three affective semantic dimensions: valence, arousal, and concreteness, and these dimensions shape their cognitive and neural processing.

Valence refers to the emotional value associated with a word, ranging from negative to positive. Arousal is the intensity of emotional activation, from calming to highly stimulating (Citron et al. 2013). Concreteness reflects the extent to which a word evokes sensory perceptible mental imagery, distinguishing tangible from abstract concepts (Walker and Hulme 1999). These dimensions are not only labels; they shape the cognitive and neural processes involved in word recognition (Wang and Yao 2012). For example, high arousal words capture early attention, and concrete words ease semantic access and engage imagery-based systems (Hofmann et al. 2009; Barber et al. 2013). Most studies have examined these attributes in isolation. Their interactions remain underexplored, especially in crossmodal and ecologically valid language contexts (Barsalou 2008).

There is debate over whether valence and arousal operate independently, with some studies proposing that each dimension contributes distinct temporal and neural signatures, whereas others suggest that they may interact dynamically. Evidence

from electroencephalography (EEG) microstates and event-related potential (ERP) studies suggests that valence and arousal exhibit dissociable neural signatures, especially in later stages such as the late positive complex (LPC), without significant interaction (Gianotti et al. 2008; Kuperman et al. 2014; Vinson et al. 2014; Delaney-Busch et al. 2016). These findings indicate that valence and arousal may engage partially independent neural and temporal processes during affective word processing. In contrast, studies examining earlier components have reported valence by arousal interactions. Citron et al. (2013) observed an interaction in the early posterior negativity (EPN), showing enhanced amplitudes for high-arousal positive words relative to low-arousal negative ones. Recio et al. (2014) identified an interaction between valence and arousal within the 275–425 ms post-stimulus window. Yao et al. (2016) extended these findings by showing that word concreteness further modulates this interaction across multiple processing stages, including the early perceptual (N170), late elaborative (LPC), and behavioral outcomes (reaction times). These conflicting findings suggest that emotional word processing is shaped not only by interactions between valence and arousal but also by broader lexical semantic factors. In particular, word concreteness has been shown to exert robust and distinct effects on the processing of emotional words.

An emerging body of research suggests that word concreteness modulates emotional word processing across distinct stages, influencing both early perceptual encoding and later evaluative processes (Yao et al. 2016; Luo and Qi 2022; Jin et al. 2023). An ERP study found that emotional valence modulated the LPC for concrete words but not abstract words, suggesting that emotional effects depend on concreteness (Kanske and Kotz 2007). This asymmetry has been attributed to the slower and less automatic semantic access of abstract words (Palazova et al. 2013). However, more recent studies challenge this binary view. ERP results show that concreteness influences both early (N170) and late (LPC) components regardless of word type (Yao et al. 2016; Pauligk et al. 2019; Luo and Qi 2022). Notably, Luo and Qi (2022) demonstrated that abstract words can evoke distinct LPC responses depending on emotional valence, indicating that abstract concepts may engage more differentiated emotional processing than previously assumed. Supporting this, fMRI evidence has shown enhanced semantic selection for emotional abstract words (Pauligk et al. 2019).

These findings point to a shift from the traditional view that abstract words are emotionally impoverished, toward the view that emotion grounds meaning, especially for abstract words. This pattern aligns with affective embodiment theory, which posits that emotion anchors abstract words in bodily states and experiential contexts, thereby compensating for their lack of direct sensory referents (Barsalou 2008; Vigliocco et al. 2014; Kousta et al. 2011).

A large body of electrophysiological research has investigated the temporal dynamics of emotional word processing, revealing distinct neural mechanisms that reflect how valence, arousal, and attention interact during lexical cognition (Citron 2012; Yao et al. 2018). Early perceptual components such as P1 are modulated by both valence and arousal. Negative words elicit larger P1 amplitudes than neutral words, reflecting heightened attentional capture by emotionally salient stimuli (Van Hooff

et al. 2008). Similarly, high-arousal words evoke greater P1 than low-arousal words (Hofmann et al. 2009), with threat-related words enhancing P1 amplitude primarily in male participants. These findings suggest that emotionally arousing stimuli trigger bottom-up attentional mechanisms at early perceptual stages. In crossmodal affective priming paradigms, emotionally incongruent auditory primes have been found to elicit enhanced P1 amplitudes relative to congruent pairings, possibly due to the additional attentional demands required to resolve mismatched affective information (Gerdes et al. 2013; Liang et al. 2025). The P1 further reflected that emotionally congruent prime–target combinations were processed more efficiently.

The EPN, typically occurring between 200 and 300 ms with occipital distribution, is consistently enhanced for emotional compared to neutral words (Palazova et al. 2011; Liu et al. 2023). The EPN is considered to index the transition from visual perception to semantic access, with arousal playing a dominant role in its modulation, consistent with Russell's (1980) circumplex model, which conceptualizes arousal and valence as orthogonal affective dimensions. Finally, the LPC (a sustained positivity emerging around 500–800 ms) reflects elaborative semantic processing and memory encoding for emotional stimuli (Kissler et al. 2007). LPC amplitudes are larger for positive and negative emotional words than for neutral words. This pattern is interpreted as engagement of cognitive resources at late processing stages that depend on valence (Herbert et al. 2008; Yao et al. 2016).

While traditional ERP analyses focus on amplitude changes in predefined components, representational similarity analysis (RSA) offers a complementary approach by examining the neural representational geometry underlying cognitive processes. RSA quantifies the similarity between spatiotemporal patterns of brain activity elicited by different stimuli or conditions, capturing the distributed coding of semantic and affective features (Kriegeskorte et al. 2008). Univariate ERP analyses rely on component peaks and latencies. RSA preserves the multivariate structure of the EEG across channels and time points, allowing tracking of how subtle stimulus differences are encoded over time (Cichy et al. 2014; Grootswagers et al. 2017; Zhang et al. 2023). RSA has been increasingly applied to EEG decoding in affective neuroscience (e.g., Li et al. 2022; Liang et al. 2025). Li et al. (2022) demonstrated that arousal-related neural patterns emerge earlier than valence patterns during emotional face perception, highlighting the temporal sensitivity of RSA in resolving competing representational dimensions. This approach moves beyond categorical contrasts such as abstract versus concrete. It models concreteness as a continuous dimension in semantic space and captures subtle gradations in neural similarity. RSA provides a way to model emotional meaning construction as a temporally distributed and multivariate neural process.

We used a crossmodal priming paradigm to examine how emotional vocalizations and affective-semantic dimensions shape word processing. Vocal primes were either linguistic (spoken words) or nonlinguistic (sounds such as laughter or sobbing), followed by emotional target words that were positive, negative, or neutral. We manipulated the emotional congruence between the prime and target, and recorded the EEG while participants performed a valence judgment task. We combined ERP analysis

with time-resolved RSA to capture both component-based and distributed neural dynamics. Guided by previous findings and the structure of our paradigm, we proposed four hypotheses.

(1) Emotional valence was expected to influence behavior and neural activity. We predicted that negative words would lead to longer reaction times than positive or neutral words. This delay has been linked in earlier studies to greater cognitive demands or to interference from avoidance tendencies (Kousta et al. 2009; Yao and Wang 2014). At the ERP level, we predicted that incongruent prime–target pairs would elicit larger P1 and EPN amplitudes. Such effects would reflect increased attentional allocation during early perceptual and semantic stages (Gerdes et al. 2013; Liang et al. 2025). (2) Arousal and valence were expected to show distinct representational dynamics in the RSA results. We predicted that arousal-related neural patterns would appear earlier, around 200 ms after target onset. Valence-related representations were expected to arise later and last longer. This pattern would be consistent with a sequential encoding of affective properties reported in earlier work (Citron et al. 2013; Yao et al. 2016; Li et al. 2022). (3) Word concreteness was expected to show sustained representational encoding across time, consistent with its semantic grounding function. RSA results were predicted to capture the gradual and stable differentiation of concreteness from about 200 ms onward (Luo and Qi 2022; Kousta et al. 2011; Vigliocco et al. 2014). (4) Vocalization type was expected to affect early neural activity. We predicted that RSA would reveal a clear distinction between linguistic and nonlinguistic vocalizations as early as –100 ms relative to target onset. These differences reflected early sensory and attentional processes linked to voice type (Pell et al. 2015; Brück et al. 2011). The design was intended to characterize how multimodal emotional cues and lexical semantic dimensions jointly shape the temporal dynamics of emotional word recognition.

2 | Materials and Procedure

2.1 | Participants

The EEG experiment recruited 31 healthy volunteers (15 males, aged 18–28 years; mean age = 22.3 years). Based on recent studies employing RSA and ERP methods to investigate word-related recognition, a sample size of 31 participants was deemed appropriate for the EEG experiment (Huang et al. 2023; Wang et al. 2024). A simulation using the Superpower R package (Lakens and Caldwell 2021) estimated that a sample size of 30 participants would be required to detect a significant interaction effect in a three-way repeated-measures ANOVA: 3 (word valence: negative, neutral, positive) × 2 (vocalization type: linguistic, nonlinguistic) × 2 (congruence: congruent, incongruent), assuming a moderate effect size, with a power of 0.8 and an α level of 0.05. Simulation-based power analysis estimates the likelihood of detecting hypothetical effects in an ANOVA design by generating simulated data from factorial designs with defined parameters. All participants were native Mandarin Chinese speakers, right-handed, and reported normal hearing and normal or corrected-to-normal vision, with no history of neurological or psychiatric disorders. Participants received a remuneration of US\$20 for their participation in the EEG experiment. The study was approved by the Ethics Committee of

Liaoning Normal University (China) and was conducted in accordance with the latest version of the Declaration of Helsinki.

2.2 | Stimuli

2.2.1 | Word Stimuli

We selected 25 Chinese adjectives for each of three emotion categories: negative (e.g., *barbarous*), positive (e.g., *kind*), and neutral (e.g., *scattered*), based on the Xinhua Dictionary, 12th edition. In total, 75 adjectives were used as experimental stimuli. To ensure balanced representation across response categories and prevent frequency-driven response bias, we included an equal number of positive, neutral, and negative words in the stimulus set. This approach is appropriate for this study's three-category valence judgment task, in which each emotional category served as a full experimental condition. Twenty participants (10 men and 10 women) who did not overlap with the EEG sample were recruited to rate the valence, arousal, and concreteness of these words using a 9-point scale (arousal: 1 = very calm, 9 = very excited; valence: 1 = very unpleasant, 9 = very pleasant; concreteness: 1 = very abstract, 9 = very concrete). Word frequency data were obtained from the Dictionary of Usage Frequency of Modern Chinese Words (Liu et al. 1990). Detailed information on each word is provided in the [Appendix](#) (Table A1). Words were included in the final stimulus set only if their average valence ratings met the following criteria: <4 for negative, 4–6 for neutral, and >6 for positive. This procedure ensured that each item conformed to its intended emotional category at the item level, rather than relying solely on group-level means. A one-way repeated-measures ANOVA was conducted to analyze valence, arousal, concreteness, number of strokes, and word log frequency. The results showed significant main effects of valence, $F(2, 72) = 3089.32$, $p < 0.001$, $\eta_p^2 = 0.998$, positive words rated significantly higher (more pleasant) in valence than neutral and negative words, and neutral words rated significantly higher than negative words. The main effect of arousal was significant, $F(2, 72) = 224.13$, $p < 0.001$, $\eta_p^2 = 0.862$, with positive and negative words rated significantly higher in arousal than neutral words, and no difference between negative and positive words. The main effect of the concreteness was not significant, $F(2, 72) = 1.32$, $p = 0.275$, $\eta_p^2 = 0.035$. There were no differences in the log frequency between the selected words, $F(2, 72) = 1.98$, $p = 0.145$, $\eta_p^2 = 0.052$. There was a significant difference in the number of strokes among the word categories, $F(2, 72) = 3.87$, $p = 0.025$, $\eta_p^2 = 0.097$, with negative and positive words containing more strokes than neutral words. These stimuli's size, background, spatial frequency, contrast, and luminance

were consistent. The size of all word stimuli was 8.6×4.3 cm (the viewing angle was $8.2^\circ \times 4.1^\circ$). More details about the descriptive statistics of word stimuli are presented in Table 1.

2.2.2 | Vocalization Stimuli

The vocalization materials used in this study have been validated and employed in previously published research, ensuring their reliability and appropriateness (Liang et al. 2025).

2.2.3 | Nonlinguistic Vocalization Stimuli

We selected 30 emotional vocalizations from the Montreal Affective Voices database (Belin et al. 2008). The set included 10 positive (e.g., *laughing*), 10 neutral (e.g., a calm “ah...”), and 10 negative (e.g., *screaming*). All vocalizations were produced by male speakers. There was no significant difference in the average vocalization duration ($M \pm SE$: 719.11 ± 0.17 ms), and all vocalizations were presented at an intensity of 60 dB SPL.

2.2.4 | Linguistic Vocalization Stimuli

We selected 30 Chinese adjectives for the experiment. Each word had two characters. The set included 10 positive (e.g., *loyal*), 10 negative (e.g., *fierce*), and 10 neutral (e.g., *abstract*). All words were taken from the Xinhua Dictionary, 12th edition. Two male speakers read each of the 30 words in a neutral tone at a natural speed, generating 60 audio recordings. Recordings were made in a quiet environment using high-quality microphones and Adobe Audition CS6, with a sampling rate of 44,100 Hz and a bit depth of 16-bit. All speakers were native Mandarin Chinese speakers and received monetary compensation for their participation. The loudness of all vocalization recordings was normalized to 60 dB SPL using Praat software. There was no significant difference in the average vocalization duration for the three emotions, $F(2, 57) = 1.35$, $p = 0.268$, $\eta_p^2 = 0.045$ ($M \pm SE$: 816.56 ± 14.28 ms).

2.2.5 | Arousal and Valence Measurements for Two Vocalizations

Seventeen healthy participants (6 males; Mean age = 22.64 years), all native Mandarin Chinese speakers who did not overlap with participants in other parts of the study, were recruited to rate the emotional attributes of two types of vocalizations. A 7-point scale was used for the ratings (arousal: 1 = very calm, 7 = very excited;

TABLE 1 | Assessment results on the three dimensions of the visual stimulus ($M \pm SD$, min–max).

Dimension	Negative	Positive	Neutral
Arousal	5.66 ± 0.57 [4.7–7.3]	5.35 ± 0.56 [4.5–6.3]	2.61 ± 0.56 [2.0–4.0]
Valence	2.07 ± 0.18 [1.6–2.3]	7.52 ± 0.21 [7.3–8]	5.08 ± 0.32 [4.6–5.8]
Concreteness	3.98 ± 0.35 [3.4–4.6]	4.05 ± 0.59 [3.3–5.5]	4.27 ± 0.93 [2.8–6.1]
Number of strokes	19.56 ± 5.39 [13–34]	19.6 ± 4.75 [11–28]	16.44 ± 3.48 [8–25]
Frequency	1.64 ± 0.58 [0.7–3.1]	1.90 ± 0.69 [0.7–3.5]	1.99 ± 0.66 [1.1–3.3]

valence: 1=very unpleasant, 7=very pleasant; see Table A2 in the Appendix for details). A two-way repeated-measures ANOVA was conducted on valence and arousal, with vocalization category (linguistic vs. nonlinguistic) and emotion (negative, positive, neutral) as within-subject factors. For valence, the results revealed a significant effect of emotion, $F(2, 32) = 284.42$, $p < 0.001$, $\eta_p^2 = 0.947$. The negative ($M \pm SE$: 2.03 ± 0.11), positive (6.00 ± 0.13), and neutral vocalizations (3.88 ± 0.08) differed significantly from each other ($ps < 0.001$). The vocalization category's main effect was insignificant, with no difference between linguistic and nonlinguistic vocalizations, $F(1, 16) = 0.01$, $p = 0.929$, $\eta_p^2 = 0.001$. For arousal, a significant main effect of emotion was found, $F(2, 32) = 77.94$, $p < 0.001$, $\eta_p^2 = 0.830$, with significantly higher ratings for positive ($M \pm SE$: 5.44 ± 0.22) and negative vocalizations (5.43 ± 0.15) than for neutral vocalizations (3.08 ± 0.29 , $ps < 0.001$). The vocalization category's main effect was insignificant, $F(1, 16) = 3.89$, $p = 0.066$, $\eta_p^2 = 0.196$. Valence and arousal ratings were collected separately for vocalization and word stimuli using modality-appropriate scales (7-point and 9-point, respectively), following conventions in the literature. These ratings were used only to categorize stimuli within each modality.

2.3 | Procedure

The experiment was conducted in a soundproof room with soft lighting. Participants sat in a comfortable chair, positioned 60cm from a 23-in. computer display. The screen had a resolution of 1920×1080 pixels and a refresh rate of 60Hz. Stimuli presentation and response recording were controlled by *E-Prime 2.0*. This experiment employed a within-subject design with 12 conditions: 3 (word valence: negative, neutral, positive) $\times$ 2 (vocalization type: linguistic, nonlinguistic) $\times$ 2 (congruence: congruent, incongruent). In the incongruent condition, the priming vocalization contained two other emotional categories (e.g., in the negative-incongruent condition, the priming vocalization conveyed positive and neutral emotions) while maintaining the same vocalization type as the incongruent stimuli. In this study, congruence refers to the alignment of emotional content within

the vocalizations and target words. More precisely, it reflects the match between emotional information conveyed by linguistic vocalizations (semantic content) or nonlinguistic vocalizations (emotional prosody) and the emotional valence of the words (see Appendix for details). The response screen was delayed during the experiment to minimize EEG noise caused by keypressing (Anwar et al. 2016).

Each trial began with a randomly presented white fixation cross (500–700 ms), followed by a 700–800 ms auditory stimulus (depending on the vocalization length) delivered through headphones. A black screen was presented for 200 ms. The emotional word was then displayed for 1000 ms. Finally, an exclamation mark appeared on the screen for 1500 ms, signaling participants to respond (Figure 1). Participants were instructed to categorize each word as positive, neutral, or negative by pressing one of three keys (spacebar, J, or K). Response button assignments (i.e., which key corresponded to which emotional category) were counterbalanced across participants to minimize potential motor or response mapping biases (Luck 2014). The practice phase consisted of 36 trials. The EEG experiment comprised three blocks with self-paced breaks in between, allowing participants to rest for as long as needed without any time limit. Each block contained 200 trials, resulting in 600 trials, with each condition appearing in 50 trials. To construct these trials, 25 unique two-character Chinese adjectives were selected for each word valence category (positive, neutral, negative). Each word appeared twice within its assigned condition only, resulting in 50 trials per condition. This design ensured that word-level repetition was fully balanced and orthogonal to the congruence manipulation, preventing any confound due to unequal word frequency or familiarity effects. Trials from all 12 experimental conditions were presented in a fully randomized order to minimize habituation effects.

2.4 | Behavioral Data Analysis

We first excluded response times (RTs) and accuracy (ACC) that exceeded three standard deviations. Trials with inaccurate

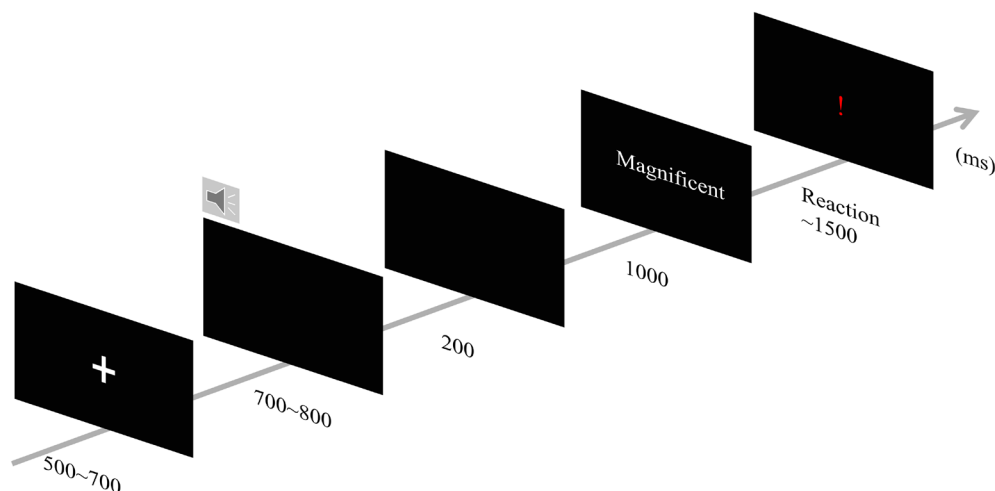


FIGURE 1 | Experimental procedure of a single trial in the EEG experiment. Each trial began with a fixation cross, followed by an auditory prime, a brief blank screen, and the presentation of a target word. Participants were instructed to make an emotional valence judgment when the response cue appeared.

responses were excluded from the RT analysis. A three-way repeated-measures ANOVA was conducted with 3 (word valence: negative, neutral, positive) $\times$ 2 (vocalization type: linguistic, nonlinguistic) $\times$ 2 (congruence: congruent, incongruent) as within-subject factors using *SPSS 20.0*. Degrees of freedom and *p* values were adjusted according to the Greenhouse–Geisser epsilon correction, and the Bonferroni correction ($p < 0.05$) was used for post hoc comparisons. As a measure of effect size, the partial eta squared (η_p^2) was reported.

2.5 | EEG Recordings and Data Analysis

2.5.1 | Preprocessing

EEG data were recorded using a 64-channel EEG system (Brain Products, Germany), with tin electrodes placed on 64 scalp sites according to the extended 10–20 system and mounted on an elastic cap. EEG signals were recorded with a 0.01–100 Hz bandpass filter and a continuous sampling rate at 500 Hz/channel. The online reference was left mastoid, and the impedances of all electrode sites were kept below 5 k Ω . ERP data were pre-processed using EEGLAB (Delorme and Makeig 2004). The raw data were first high-pass filtered at 0.1 Hz, low-pass filtered at 30 Hz, and re-referenced offline to the average. The data with large artifacts (exceeding $\pm 50 \mu\text{V}$) were removed. Components contaminated by eye blinks were detected and removed using the independent component analysis (ICA) and ICAlabel algorithm (Pion-Tonachini et al. 2019). The data were segmented into 1000 ms epochs time-locked to word stimuli onset, starting 200 ms before stimuli onset. The epochs' data were baseline corrected using the 200 ms before the stimulus onset. On average, about 6.14% of the trials per condition were rejected due to artifacts, and the rejection rate did not differ across conditions, $F(11, 348) = 0.13$, $p = 1.00$, $\eta_p^2 = 0.004$. No participants were excluded from the analysis. After excluding trials due to artifacts and incorrect responses, the mean number of valid trials per condition was $M = 44.91$ ($SD = 0.29$). Trial numbers were comparable across all conditions, ensuring sufficient signal quality for ERP averaging.

2.5.2 | ERP Analysis

Group-level ERP waveforms were computed by averaging single-participant ERP waveforms, and group-level scalp topographies at corresponding peak latencies were obtained by spline interpolation. The dominant ERP components were identified based on their characteristic latencies, scalp distributions, and functional significance, as guided by prior literature (Citron 2012; Zhang et al. 2014; Luo and Qi 2022; Liu et al. 2023). The time windows for P1 (100–140 ms), EPN (170–250 ms), and LPC (350–650 ms) were determined based on previous studies and confirmed by visual inspection of the grand-averaged waveforms and topographies in the current data. P1 amplitude was measured from PO7, PO8, P7, P8, P5, and P6 electrodes; EPN amplitude was measured from PO3, PO4, PO7, PO8, P5, P6, O1, and O2 electrodes; LPC amplitude was measured from CP1, CP2, PZ, CZ, CPZ. P1 and EPN were analyzed using a four-way repeated measures ANOVA with 3 (word valence: negative, neutral, positive) $\times$ 2

(vocalization type: linguistic, nonlinguistic) $\times$ 2 (congruence: congruent, incongruent) $\times$ 2 (hemisphere: left, right). LPC was analyzed using a three-way repeated measures ANOVA with 3 (word valence: negative, neutral, positive) $\times$ 2 (vocalization type: linguistic, nonlinguistic) $\times$ 2 (congruence: congruent, incongruent).

2.5.3 | Decoding Analysis and RSA

Multivariate time-series decoding analysis was applied to the EEG data using support vector machine (SVM) classification and functions from the *CoSMoMVPA* toolbox (Chang and Lin 2011; Oosterhof et al. 2016) to explore further how the brain encodes this sensory information. We downsampled the segmented data to 250 Hz and subsequently performed binary classification to construct a neural representational dissimilarity matrix (RDM), where the pairwise decoding accuracy served as the dissimilarity metric. SVM classifiers were trained and tested at each time point across all the electrodes. We randomly assigned each trial of one condition (a total of 12 experimental conditions) to one of the ten splits and averaged the trials in each split to improve the signal-to-noise ratio. Training and testing were implemented using the leave-one-out method to avoid double-dipping: the classifier was trained on all but one averaged trial of one condition and tested on the left-out averaged trial. In this procedure, every averaged trial was excluded once, and the accuracy was averaged across these repetitions. To obtain decoding accuracies, we performed the binary classification on all possible pairwise comparisons between conditions (66 pairs, calculated as $(12-1) \times 12/2$). All steps were repeated 100 times, and the accuracies over repetitions were averaged to construct a 12×12 neural RDM for each subject at each time point (300 neural RDMs per subject).

In addition, we constructed arousal RDM, valence RDM, and concreteness RDM from the material measurement data of words by calculating the Euclidean distance between words to indicate their dissimilarity. Words were selected based on independent valence, arousal, and concreteness ratings to ensure sufficiently wide and representative distributions for RSA (details in Table 1). Each emotional category preserved internal variance on both dimensions, supporting their use as continuous predictors in RSA modeling. We constructed vocalization RDM according to vocalization type, with the dissimilarity as 0 if it is the same type of vocalization and 1 otherwise (Figure 2). We calculated the partial correlation between the different model RDMs and the neural RDMs at each time point for each subject. As some models were correlated (e.g., arousal, valence, and concreteness), we could separate the contribution of these models from other models after excluding them (Giordano et al. 2013; Cichy and Pantazis 2017). These partial correlation coefficients can be used as indicators of the time course of the different dimensions in the data.

2.6 | Searchlight Analysis

We ran a multilevel regression-based searchlight RSA across the time points to investigate which specific electrodes contributed to the relationship between neural activity and the four RDMs. For

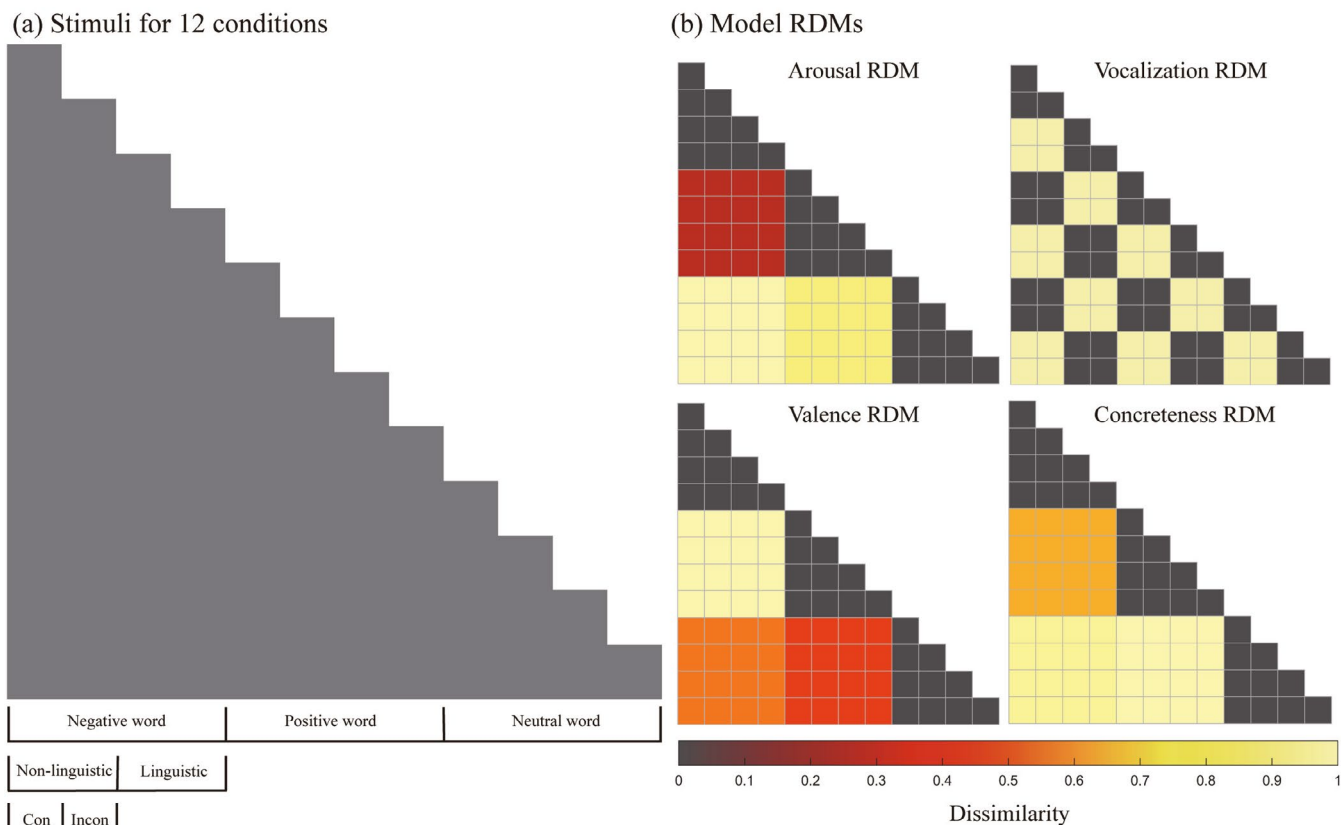


FIGURE 2 | RSA model structure. (a) The arrangement of the 12 experimental conditions used in the RSA analysis. (b) Model representational dissimilarity matrices (RDMs) for each stimulus dimension. Con, congruent; Incon, incongruent.

each electrode, we constructed a neighborhood centered on the electrode and the four nearest electrodes around it. At each time point, the response pattern for each of the 12 conditions within the neighborhood was vectorized, and the neural phase anisotropy of each category pair was calculated using the squared Euclidean distance to produce a 12×12 neural RDM. In other words, it was constructed based on the amplitude of each neighborhood rather than the decoding accuracy. In addition, a general linear regression (GLM) was used to build the neural RDM from the four model RDMs for each neighborhood across time points. The regression coefficients of the four model RDMs were mapped to the central electrodes of the neighborhoods. The scalp distribution of the regression coefficients describes the whole-brain correspondence between the neural and model RDMs.

2.7 | Temporal Generalization Analysis

We employed the temporal generalization approach of multivariate pattern analysis (MVPA) (King and Dehaene 2014) to investigate whether different dimension representations were stable or changed over time. This approach trains a classifier on data at a specific time point (training time). It tests it at all other time points (generalization time) to investigate whether the representation remains stable. If the classifier can distinguish between two conditions at the trained time and later time points, it indicates a stable representation format. Otherwise, it suggests a change in the format. We conducted temporal generalization analyses for each participant

and three dimensions (all conditions, word valence, and vocalization type) and then averaged the decoding accuracies of each subject. The analysis resulted in three 300×300 matrices (-200 – 1000 ms to stimulus onset) that captured the classifier's generalization performance across time.

2.8 | Statistical Inference

We performed permutation-based cluster size inference to examine the time periods when the decoding accuracy or correlation showed significant effects (Nichols and Holmes 2002). The null hypothesis corresponded to a correlation coefficient of zero and a decoding accuracy of 50%. Significant temporal clusters were specified as adjacent time points that exceeded a statistical boundary line (i.e., cluster-inducing threshold). The condition labels of the data were permuted by random multiplication by $+1$ or -1 (i.e., the sign permutation test), and this process was repeated 5000 times to obtain the t -value distribution of all possible permutations. Based on the t -value distribution, the boundary line was determined using a sign-permutation test. At each time point, the 95th percentile of the t -value distribution was used as the clustering induction threshold (equivalent to $p < 0.05$, one-sided), and the 95th percentile of the maximum cluster size among all permutations was used as the significant cluster. This approach provided us with significant periods for decoding accuracy and correlation. In the searchlight analysis, we first located the significant time points of each channel and then marked the significant channels on the topographic map at each time point.

2.9 | Determination of Confidence Intervals (CIs)

Bootstrap tests estimated the statistical differences and 95% confidence intervals (CIs) for the onset, least, peak, and duration latencies between dimensions. The correlation coefficient was bootstrapped 1000 times, resulting in an empirical distribution of the onset (the earliest significant time point), peak (the significant time point at the maximum value), offset (the last significant time point), and duration latencies (the length of the significant period). The 95% CIs for these latencies were defined as the 2.5th and 97.5th percentiles of the distribution. Latency differences were then calculated between each pair of dimensions in these 1000 bootstrap samples, resulting in an empirical distribution of latency differences. The p -value was defined as the proportion of samples with differences greater or smaller than zero, divided by the number of permutations (i.e., a two-sided test), and was corrected for multiple comparisons using a false discovery rate (FDR) of 0.05.

3 | Results

All descriptive behavioral and ERP data results are expressed as $M \pm SE$.

3.1 | Behavioral Results

3.1.1 | ACC

The main effect of word valence was not significant, $F(2, 58) = 3.13$, $p = 0.051$, $\eta_p^2 = 0.097$ (negative = $97\% \pm 0.7\%$, positive = $96.6\% \pm 0.6\%$, neutral = $96.1\% \pm 0.8\%$). The main effect of vocalization type was not significant, $F(1, 29) = 1.31$, $p = 0.261$, $\eta_p^2 = 0.043$ (nonlinguistic = $96.4\% \pm 0.7\%$, linguistic = $96.7\% \pm 0.7\%$). The main effect

of congruence was not significant, $F(1, 29) = 2.63$, $p = 0.116$, $\eta_p^2 = 0.083$ (congruent = $96.4\% \pm 0.7\%$, incongruent = $96.8\% \pm 0.7\%$). No significant interaction effects were observed.

3.1.2 | RT

The main effect of word valence was significant, $F(2, 58) = 5.88$, $p = 0.005$, $\eta_p^2 = 0.169$, with significantly longer RTs for negative words than positive and neutral words ($ps < 0.001$), and no difference between positive and neutral words (negative = 285.36 ± 13.37 ms, positive = 274.82 ± 12.54 ms, neutral = 274.82 ± 13.03 ms). The main effect of vocalization type was not significant, $F(1, 29) = 0.17$, $p = 0.686$, $\eta_p^2 = 0.006$ (nonlinguistic = 278.68 ± 12.96 ms, linguistic = 277.99 ± 12.74 ms). The main effect of congruence was not significant, $F(1, 29) = 0.23$, $p = 0.634$, $\eta_p^2 = 0.008$ (congruent = 278.89 ± 12.87 ms, incongruent = 277.78 ± 12.88 ms). No significant interaction effects were observed.

3.2 | ERP Results

The ERP analysis examined the time-locked neural responses to emotional words as modulated by prime congruence, vocalization type, and word valence. Specifically, we focused on early perceptual (P1), semantic (EPN), and late evaluative (LPC) components to capture distinct stages of emotional word processing.

3.3 | P1

P1 results showed a significant main effect of congruence (Figure 3a), $F(1, 29) = 43.59$, $p < 0.001$, $\eta_p^2 = 0.601$, with the incongruent condition eliciting significantly greater P1 amplitudes

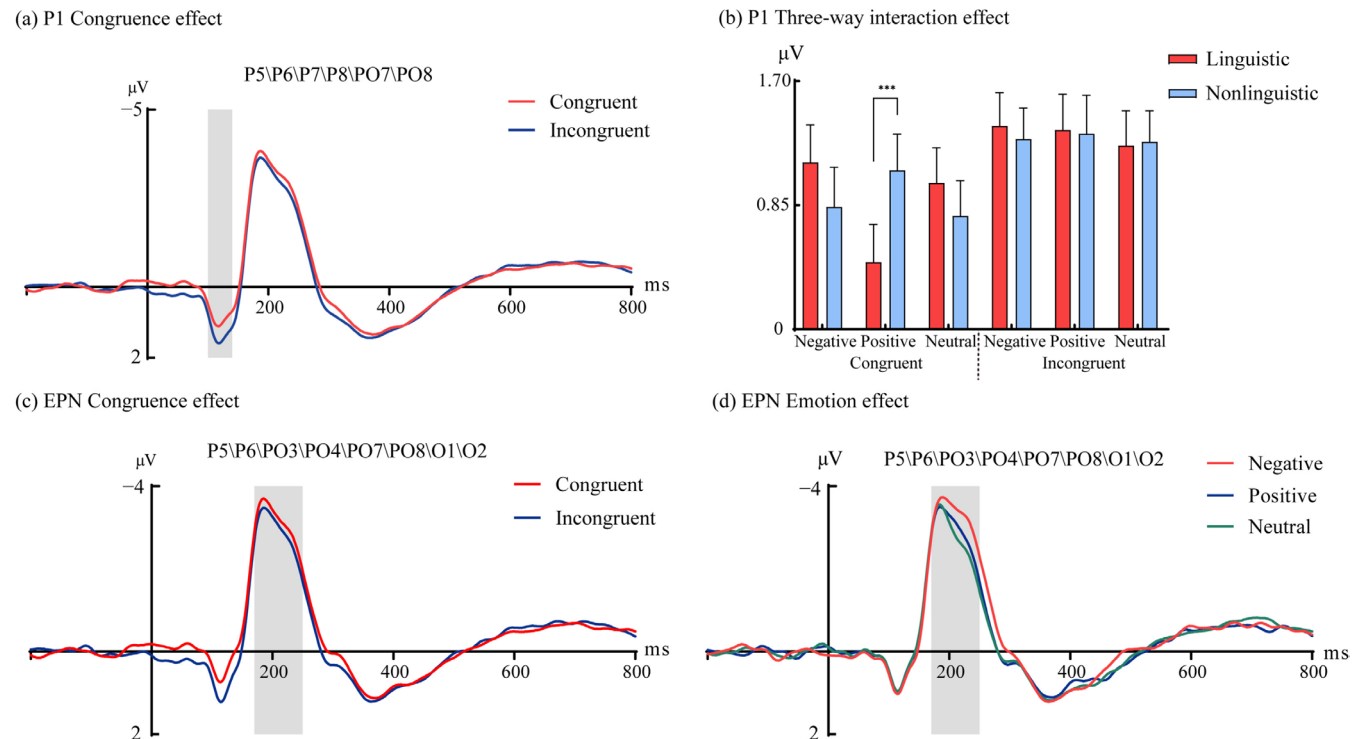


FIGURE 3 | P1 and EPN results. Error bars represent the standard error of the mean (SEM). (a) P1 congruence effect. (b) P1 three-way interaction effect. (c) EPN congruence effect. (d) EPN emotion effect. Asterisks denote statistical significance: *** $p < 0.001$.

than the congruent condition (congruent = $0.88 \pm 0.22 \mu\text{V}$, incongruent = $1.32 \pm 0.21 \mu\text{V}$). The main effect of word valence was not significant, $F(2, 58) = 0.83$, $p = 0.439$, $\eta_p^2 = 0.028$ (negative = $1.17 \pm 0.22 \mu\text{V}$, positive = $1.06 \pm 0.23 \mu\text{V}$, neutral = $1.08 \pm 0.21 \mu\text{V}$). The main effect of vocalization type was not significant, $F(1, 29) = 0.00$, $p = 0.992$, $\eta_p^2 = 0.000$ (linguistic = $1.10 \pm 0.22 \mu\text{V}$, nonlinguistic = $1.10 \pm 0.21 \mu\text{V}$). The three-way interaction effect of word valence $\times$ vocalization type $\times$ congruence was significant, $F(2, 58) = 3.48$, $p = 0.037$, $\eta_p^2 = 0.107$. Post hoc tests revealed that under congruent conditions, linguistic vocalizations elicited larger P1 amplitudes than nonlinguistic vocalizations when participants identified positive words (positive-linguistic-congruent = $1.09 \pm 0.25 \mu\text{V}$, positive-nonlinguistic-congruent = $0.46 \pm 0.26 \mu\text{V}$, $p = 0.004$; Figure 3b).

3.4 | EPN

EPN results showed a significant main effect of congruence, $F(1, 29) = 8.00$, $p = 0.008$, $\eta_p^2 = 0.216$, with larger EPN amplitudes under the congruent condition compared with the incongruent condition (congruent = $-2.723 \pm 0.435 \mu\text{V}$, incongruent = $-2.508 \pm 0.435 \mu\text{V}$, Figure 3c). The main effect of word valence was significant, $F(2, 58) = 13.88$, $p < 0.001$, $\eta_p^2 = 0.324$, with negative words eliciting larger EPN amplitude than positive and neutral words ($ps < 0.05$), and no significant difference between positive and neutral words (negative = $-2.889 \pm 0.455 \mu\text{V}$, positive = $-2.531 \pm 0.443 \mu\text{V}$, neutral = $-2.426 \pm 0.411 \mu\text{V}$, Figure 3d). The main effect of vocalization type was not significant, $F(1, 29) = 0.32$, $p < 0.575$, $\eta_p^2 = 0.011$ (linguistic = $-2.63 \pm 0.42 \mu\text{V}$, nonlinguistic = $-2.60 \pm 0.45 \mu\text{V}$). No significant interaction effects were observed.

3.5 | LPC

The main effect of word valence was not significant, $F(2, 58) = 0.46$, $p = 0.634$, $\eta_p^2 = 0.016$ (negative = $1.54 \pm 0.20 \mu\text{V}$, positive = $1.58 \pm 0.21 \mu\text{V}$, neutral = $1.51 \pm 0.22 \mu\text{V}$). The main effect of vocalization type was not significant, $F(1, 29) = 0.54$, $p = 0.469$, $\eta_p^2 = 0.018$ (linguistic = $1.56 \pm 0.21 \mu\text{V}$, nonlinguistic = $1.54 \pm 0.21 \mu\text{V}$). The main effect of congruence was not significant, $F(1, 29) = 0.14$, $p = 0.713$, $\eta_p^2 = 0.005$ (congruent = $1.55 \pm 0.21 \mu\text{V}$, incongruent = $1.54 \pm 0.21 \mu\text{V}$). No significant interaction effects were observed.

3.6 | Decoding Results and RSA Results

The decoding and RSA analyses investigated the representational dynamics of different lexical and emotional dimensions over time. These methods allowed us to go beyond traditional ERP peak analyses by identifying when and how specific attributes such as valence, arousal, concreteness, and vocalization type were encoded in neural patterns. SVM classifiers were trained and tested on each pair of conditions at each time point across all channels for each participant. Decoding accuracy was averaged across all pairs at each time point to generate a time-resolved accuracy curve (Figure A1). The results showed that the 12 conditions could be significantly distinguished beginning at 52 ms, $t(29) = 2.51$, $p < 0.05$, Cohen's $d = 0.93$, cluster-corrected

sign permutation test (cluster-defining threshold: $p < 0.05$; cluster-corrected significance level: $p < 0.05$). The mean decoding accuracy was 52.74%, peaking at 244 ms and decreasing after 724 ms (see Appendix for details). These results show how EEG responses are discriminated at the level between the 12 conditions and provide an important basis for the following RSA.

RSA results revealed the time course of arousal (204–568 ms, 656–728 ms), valence (196–304 ms, 512–920 ms), vocalization type (0–80 ms, 88–128 ms), and word concreteness (188–996 ms), all determined using a cluster-corrected sign permutation test (cluster-defining threshold: $p < 0.05$; cluster-corrected significance level: $p < 0.05$; Figure 4a). The vocalization type was first represented from the EEG signal representation, and then word concreteness information was extracted at approximately 196.9 ms. Arousal was extracted at 218.6 ms, and valence information was finally extracted at about 252.7 ms (Figure 4b; all $ps < 0.001$, two-sided bootstrap test, FDR corrected). Compared with arousal, valence, and concreteness, vocalization type had significantly earlier onsets and shorter durations ($ps < 0.001$, two-sided bootstrap test, FDR corrected), indicating a rapid and immediate impact on word processing.

Furthermore, concreteness exhibited a significantly longer duration than arousal ($p < 0.01$) and valence ($p < 0.05$), suggesting that concreteness exerts a sustained influence on emotional word processing. These suggest that concreteness may have a sustained impact on word processing. Finally, arousal peaked significantly earlier (≈ 475.6 ms) than valence and concreteness ($ps < 0.001$, two-sided bootstrap test, FDR corrected), whereas valence peaked latest (≈ 639.5 ms). These results suggest that, during early-stage processing, cognitive resources are preferentially allocated to arousal. Valence serves as a key reference in later-stage meaning assessment, and concreteness contributes throughout the entire course of lexical cognition.

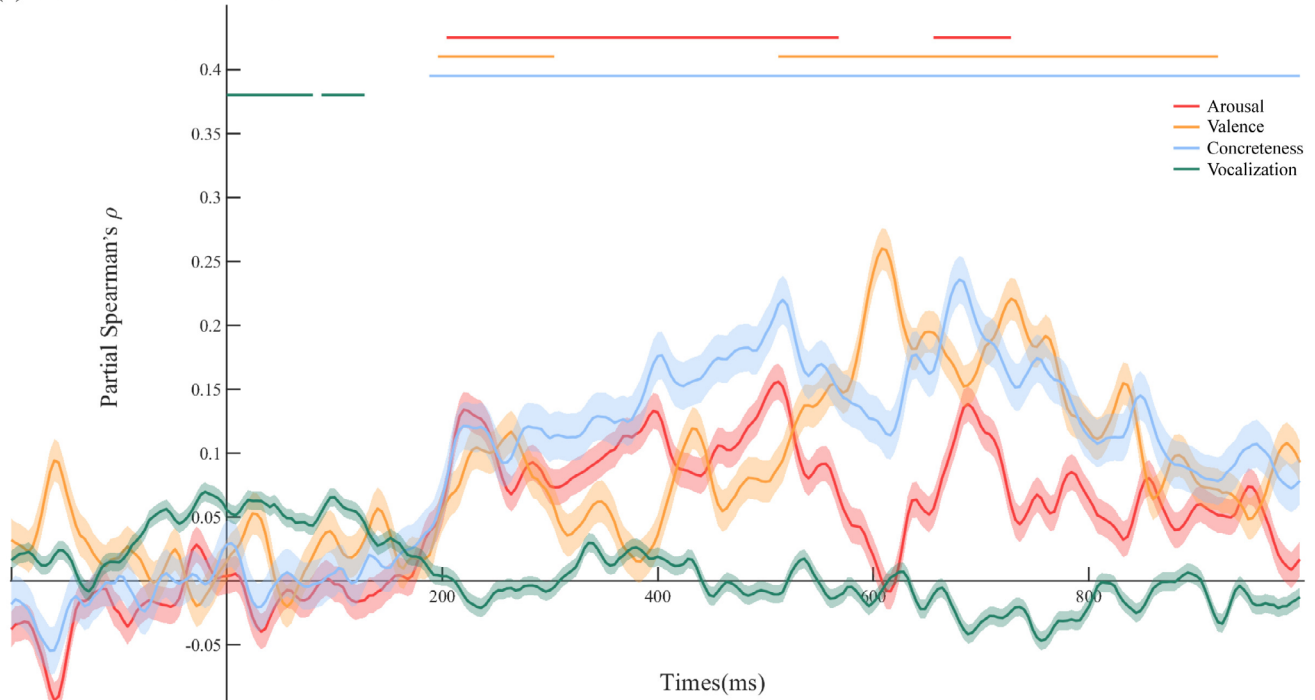
3.7 | Searchlight Analysis

The searchlight analysis was conducted to localize the spatial distribution of brain activity associated with each dimension of interest. By examining the contribution of individual electrode clusters, this approach allowed exploration of the topographic specificity of lexical and affective representations. Topographies at significant time points were identified using bootstrap tests (Figure 5). Significant channels for vocalization type were observed in the bilateral temporal, occipital, and parietal regions. Significant effects were observed predominantly in parietal and occipital regions for arousal. For valence, significant channels were distributed across parietal and frontal regions. For concreteness, significant effects were observed in frontoparietal and parietal regions.

3.8 | Temporal Generalization Analysis

Temporal generalization analysis was conducted to evaluate the stability of neural representations across processing stages. This method examined whether the encoding format for each dimension remained consistent over time or shifted during word recognition. For the condition dimension, generalization was

(a) RSA time course result



(b) Bootstrap tests results

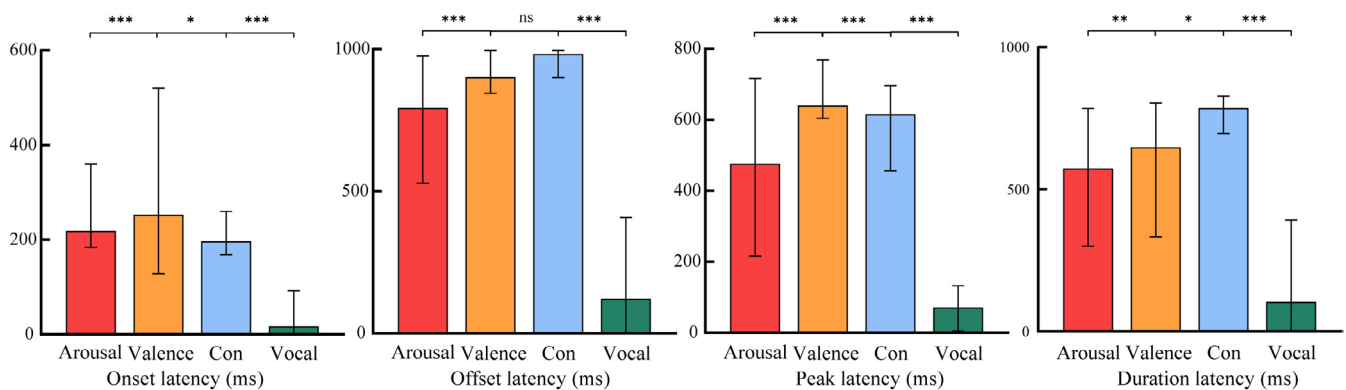


FIGURE 4 | RSA results. (a) The time course of partial Spearman correlations between neural RDMs and model RDMs is shown for arousal (red), valence (orange), concreteness (light blue), and sound (green). The lines above the horizontal axis indicate significant time points using the cluster-corrected sign reciprocity test, and the line colors represent the respective dimensions (cluster definition threshold $p < 0.05$, cluster-corrected significance level $p < 0.05$). (b) Onset, least, peak, and duration latencies for 4 model RDMs. Error bars indicate bootstrapped 95% confidence intervals. Statistical significance is indicated as follows: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; ns = not significant ($p > 0.05$). All tests were two-sided and FDR-corrected. Con, concreteness; Vocal, vocalization.

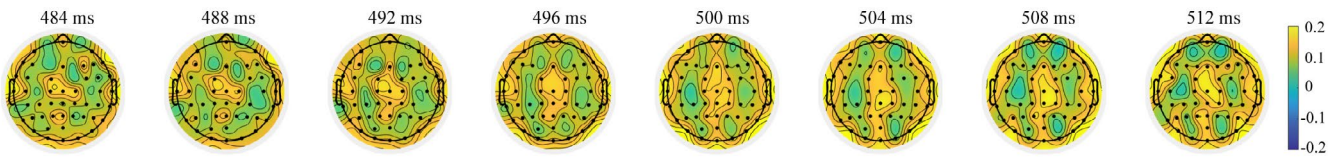
observed from approximately 220 to 700 ms (Figure 6). The narrow generalization band indicates that neural representations were transient and interpretable primarily across adjacent time points. In contrast, the emotion dimension exhibited sustained generalization from approximately 200 to 1000 ms. The generalization matrix displayed two distinct temporal clusters, with an apparent shift in the pattern emerging around 500 ms. This pattern may reflect a transition from early perceptual encoding to later evaluative stages of emotional processing; however, because temporal generalization values are not independent across time points and no established method can statistically confirm a precise boundary, this observation was interpreted descriptively rather than tested statistically. For the vocalization dimension, generalization was observed from approximately

–100 to 200 ms. These results indicate that the neural system could reliably differentiate between linguistic and nonlinguistic vocalizations even before the onset of the visual target. The persistence of this distinction into early post-stimulus periods indicates that vocalization type information may contribute to preparatory or early integrative processes during emotional word recognition.

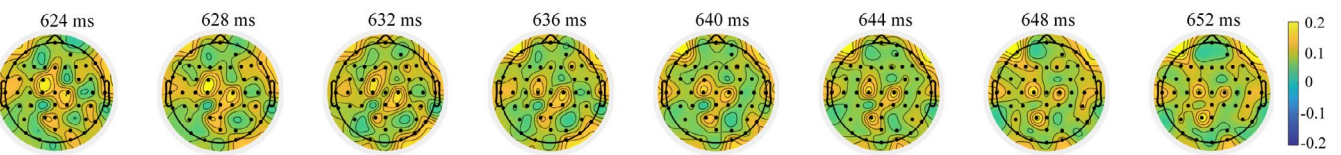
4 | Discussion

The present study investigated how linguistic and nonlinguistic emotional vocalizations modulate emotional word recognition and how three affective-semantic dimensions (valence, arousal,

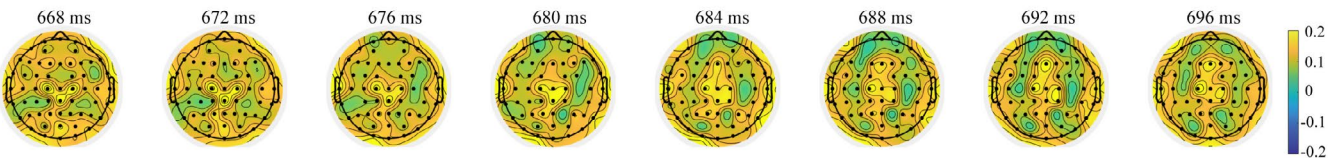
Arousal



Valence



Concreteness



Vocalization

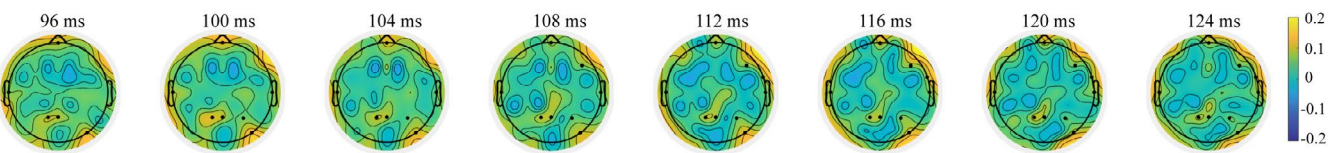


FIGURE 5 | Topographic map of searchlight RSA results. The plots display beta values from the general linear model (GLM) as a topographic distribution. Black dots indicate EEG channels that reached statistical significance ($p < 0.05$, cluster-corrected sign permutation test).

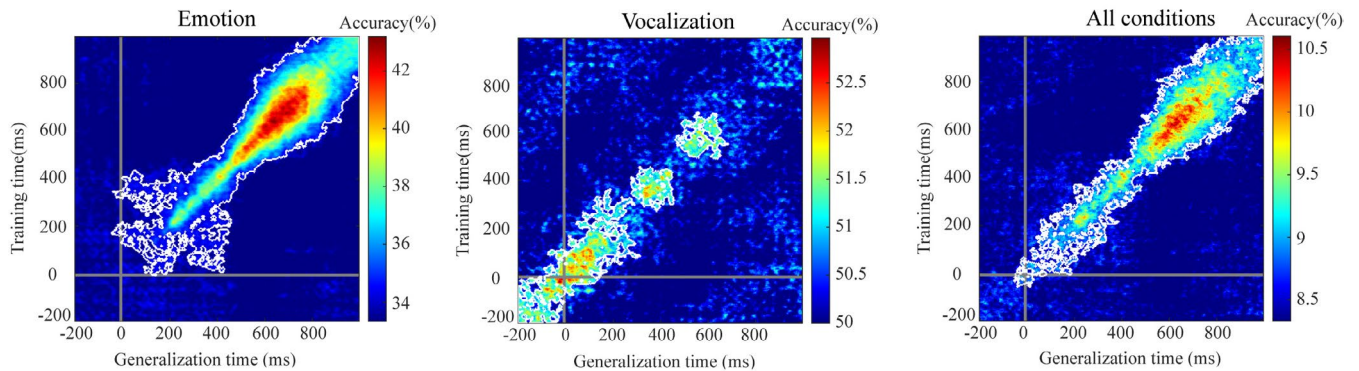


FIGURE 6 | Temporal generalization results. This figure presents the temporal cross-classification matrices for different stimulus dimensions. The y-axis indicates the training time relative to stimulus onset, and the x-axis represents the corresponding generalization time. The color scale reflects the classification accuracy for each training and testing time point combination. White outlines denote statistically significant clusters ($p < 0.05$, cluster-corrected sign permutation test).

and concreteness) unfold over time during word processing. Using a crossmodal priming paradigm, ERP analysis was combined with time-resolved RSA to capture both component-based and distributed neural dynamics. It was hypothesized that emotional congruence between the vocal prime and the visual target would modulate early perceptual and semantic processing, with the direction of modulation potentially differing across stages (P1 and EPN). Vocalization type would influence early-stage neural activity, with potential differences detectable in RSA results. Neural patterns associated with arousal would emerge earlier than those associated with valence, whereas concreteness would exhibit sustained representation across processing

stages. The results largely supported these predictions, revealing distinct temporal signatures for each dimension and highlighting the role of crossmodal priming in shaping auditory and visual interactions in word recognition. These findings contribute to theoretical models of affective language processing by combining ERP and RSA evidence within a unified temporal framework.

The P1 analysis showed a main effect of emotional congruence. Incongruent prime–target pairs produced larger amplitudes than congruent pairs, consistent with an early mismatch cost in crossmodal processing (Gerdes et al. 2013; Armitage

and Eerola 2022). In addition, a three-way interaction was found. Under congruent conditions, nonlinguistic vocalizations evoked larger P1 amplitudes than linguistic ones when participants processed positive target words. There was no overall effect of vocalization type, suggesting that this early perceptual modulation depended on context rather than being a general property of nonlinguistic stimuli (Belin et al. 2008; Ethofer et al. 2009). The fact that effects of vocalization type appeared only in specific emotional contexts highlights the need to consider both vocalization type and congruence when interpreting early ERP modulations (Pell et al. 2015; Kotz and Paulmann 2007; Brück et al. 2011). Within the crossmodal priming framework, the P1 effect mainly reflects the mismatch cost between prime and target, rather than a general response to vocalization type.

Beyond the perceptual stage indexed by P1, the EPN showed two main effects. Congruent pairs elicited larger amplitudes than incongruent ones, and negative words produced larger amplitudes than positive and neutral words. These findings suggest that the EPN reflects prioritized access to emotionally salient lexical and semantic information (Kissler et al. 2009; Schacht and Sommer 2009), and also sensitivity to emotional congruence in affective priming (Kissler et al. 2007; Hofmann et al. 2009; Gerdes et al. 2013). These findings are consistent with recent work showing that emotional congruence between context and target modulates mid-latency ERPs in language tasks, while complementary evidence indicates that the EPN reliably indexes emotional word processing in Chinese materials across different task demands (Gu et al. 2023; Liu et al. 2023).

No significant effects were found for the LPC, which is usually linked to elaborative semantic evaluation and memory-related processes (Kissler et al. 2007; Herbert et al. 2008). Previous studies have shown that emotional ERP effects are task dependent and may be reduced under explicit evaluative demands (Schacht and Sommer 2009; Citron 2012; Delaney-Busch et al. 2016). In the current study, the short stimulus onset asynchrony and the requirement of an explicit valence judgment may have limited extended evaluative processes. This limitation likely reduced the chance of LPC modulation. Although univariate P1 and EPN amplitudes did not separate linguistic from nonlinguistic vocalizations, the time-resolved RSA revealed distinct representational patterns for vocalization type during partly overlapping time windows. The RSA results demonstrate the value of multivariate methods in detecting distributed codes that may not be visible in ERP amplitudes (Kriegeskorte et al. 2008; Cichy and Pantazis 2017; Weber et al. 2024).

The RSA results showed that vocalization type was encoded as early as and only briefly in the EEG signal, with significant model and neural correlations around target onset (0–128 ms). This finding was consistent with the temporal generalization result, which indicated that the distinction was already prepared before the visual word appeared (about –100 to 200 ms). Early representational patterns related to vocalization primes were detectable during subsequent visual processing (Kriegeskorte et al. 2008; Grootswagers et al. 2017). While such effects did not emerge in component-level ERP analyses, they were observable in distributed representational analyses. The present findings highlight the sensitivity of time-resolved RSA in detecting

implicit codes (Cichy and Pantazis 2017; Weber et al. 2024). The early and short-lived encoding of vocalization type is also in line with recent time-resolved RSA findings showing that voice-related attributes are decoded quickly in EEG (Lavan et al. 2024), offering a possible route through which vocal primes can influence word recognition across modalities.

Shifting from categorical effects of vocalization type to continuous affective-semantic dimensions, the RSA results revealed distinct temporal trajectories for these dimensions. Arousal representations emerged earlier, about 204–568 ms, and peaked sooner than valence. This pattern is consistent with the view that arousal supports rapid salience-based prioritization, whereas valence contributes to later evaluative processing (LeDoux 2012; Citron et al. 2013; Delaney-Busch et al. 2016). Valence showed a later onset and a more sustained pattern (about 512–920 ms), which aligns with its proposed role in meaning evaluation and decision-related processes. Concreteness exhibited the longest continuous representational span, approximately 188–996 ms, and remained significant after controlling for arousal and valence. The evidence implies that concreteness serves as a foundational semantic dimension rather than merely a correlate of emotionality. Its sustained footprint accords with theories that view the concrete–abstract distinction as a scaffold for lexical access through mental imagery and experiential codes. Specifically, more concrete words are assumed to activate perceptual and sensorimotor codes grounded in direct experience, while more abstract words depend more on linguistic or associative links (Paivio 1986; Barsalou 2008; Yao et al. 2016). Current results also align with recent evidence showing that concreteness produces robust representational structures across tasks (Montefinese et al. 2025) and with multivariate demonstrations of implicit semantic coding in EEG (Weber et al. 2024). Together with the ERP results, these patterns point to a layered architecture of word processing. In this framework, auditory context (vocalization type) is encoded first, arousal drives early prioritization, valence shapes later evaluative stages, and concreteness offers a stable representational backbone across processing stages (King and Dehaene 2014; Li et al. 2022).

Building on the RSA evidence for distinct temporal profiles of arousal, valence, and concreteness, the ERP and RSA results suggest a layered architecture of emotional word recognition under crossmodal priming. The auditory context, reflected as an early and short-lived code for vocalization type, emerged before target onset and extended into the earliest post-onset interval, indicating that voice-derived attributes are rapidly decoded in the brain and influence subsequent perceptual stages (Lavan et al. 2024; Liang et al. 2025). Emotional congruence influenced perceptual (P1) and early semantic (EPN) processing. These findings suggest that information flowed more smoothly beyond the first encoding stage when the prime and target matched in affective meaning. Within this framework, arousal elicited earlier and short-lived representations, while valence was associated with later and more enduring codes linked to evaluative meaning. The RSA evidence suggests that such meaning-related codes may be pre-activated and gradually refined (Wang et al. 2024; Angulo-Chavira et al. 2025). Concreteness exhibited the most persistent representational signature, indicating a stable foundation that interacts with affective dimensions across processing stages and aligns with recent evidence showing its

role as an organizing principle of lexical representations (Kousta et al. 2011; Botch et al. 2024; Montefinese et al. 2025).

These results help reconcile conflicting findings by showing that different dimensions dominate different processing windows within an anticipatory, voice-derived context. This framework captures this sequence well, which combines crossmodal vocal priming with time-resolved RSA (Weber et al. 2024). Beyond theory, these dynamics have practical implications for multimodal communication. Pairing text with emotionally congruent vocal cues, such as in education, clinical assessment, or voice interfaces, may facilitate early visual processing (Campanella 2023). Incongruent cues may impose additional perceptual demands. By showing that arousal and valence make sequential, stage-specific contributions on top of a concreteness framework, the findings also help integrate prior debates about which dimension drives emotional word effects (Bayer et al. 2010; Delaney-Busch et al. 2016; Schindler et al. 2023; Botch et al. 2024).

Several constraints should be acknowledged when interpreting the present findings. First, although we controlled key lexical variables, the stimuli were not fully balanced in stroke count, which is known to influence orthographic recognition in Chinese (Tong et al. 2009). In the present study, neutral words had slightly fewer strokes than negative and positive words, but the differences were modest with large standard deviations. Since stroke count was not systematically linked to emotion category, any residual effect was likely confined to early perceptual stages (e.g., P1) and unlikely to extend to later processing. Second, our vocal stimuli were recorded from male speakers only, which precluded testing gender-related modulation in emotional voice processing. Prior work suggests that talker gender can shape affective responses, underscoring the need for multi-speaker, gender-balanced designs in future research (Chun et al. 2012). Third, the explicit valence judgment task and short stimulus onset asynchrony likely shaped evaluative engagement and may have attenuated LPC, consistent with task-dependent emotional ERP effects reported previously (Schacht and Sommer 2009; Citron 2012). Finally, while searchlight RSA helped reveal distributed representations, this analysis lacked anatomical precision and should be interpreted as exploratory. Future studies could combine RSA with source reconstruction or fMRI to provide stronger anatomical inferences (Cichy and Pantazis 2017).

5 | Conclusion

The present study employed a crossmodal priming paradigm and combined ERP with time-resolved RSA to examine how linguistic and nonlinguistic emotional vocalizations shape emotional word recognition. ERP results indicated that emotional congruence modulated both early perceptual (P1) and semantic (EPN) stages, and that vocalization type exerted context-dependent effects during these stages. RSA revealed early and transient representations of vocalization type, a temporal dissociation between arousal and valence, and a sustained contribution of concreteness across processing stages. These findings clarify the temporal architecture of affective-semantic processing and provide a methodological framework for probing distributed neural

dynamics in multimodal communication. Moreover, these results offer an integrated understanding of how arousal, valence, and concreteness interact over time, thereby reconciling the conflicting views on the relative contributions of arousal and valence in emotional word processing.

Author Contributions

Junyu Liang: conceptualization, methodology, formal analysis, visualization, writing – original draft, writing – review and editing, data curation, project administration. **Mingming Zhang:** conceptualization, writing – review and editing, supervision. **Lan Yang:** writing – review and editing. **Hongmin Luo:** writing – review and editing. **Hongying Li:** writing – review and editing. **Jun Chen:** supervision, resources, writing – original draft, writing – review and editing. **Wenbo Luo:** supervision, resources, conceptualization, writing – review and editing, writing – original draft, funding acquisition.

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Ethics Statement

All procedures performed in this study followed the 1964 Helsinki Declaration and its later amendments or comparable ethical standards.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

All data generated/analyzed during this study is available from the corresponding author upon reasonable request.

References

- Angulo-Chavira, A. Q., A. M. Castellón-Flores, H. Carrasco-Ortiz, and N. Arias-Trejo. 2025. “Hyper-Speed Meaning and Form Predictions: An EEG-Based Representational Similarity Analysis.” *Psychonomic Bulletin & Review*. <https://doi.org/10.3758/s13423-025-02731-4>.
- Anwar, A. R., M. Muthalib, S. Perrey, et al. 2016. “Effective Connectivity of Cortical Sensorimotor Networks During Finger Movement Tasks: A Simultaneous fNIRS, fMRI, EEG Study.” *Brain Topography* 29, no. 5: 645–660.
- Armitage, J., and T. Eerola. 2022. “Cross-Modal Transfer of Valence or Arousal From Music to Word Targets in Affective Priming?” *Auditory Perception & Cognition* 5, no. 3–4: 192–210.
- Banse, R., and K. R. Scherer. 1996. “Acoustic Profiles in Vocal Emotion Expression.” *Journal of Personality and Social Psychology* 70, no. 3: 614–636.
- Barber, H. A., L. J. Otten, S. T. Kousta, and G. Vigliocco. 2013. “Concreteness in Word Processing: ERP and Behavioral Effects in a Lexical Decision Task.” *Brain and Language* 125, no. 1: 47–53.
- Barsalou, L. W. 2008. “Grounded Cognition.” *Annual Review of Psychology* 59, no. 1: 617–645.
- Bayer, M., W. Sommer, and A. Schacht. 2010. “Reading Emotional Words Within Sentences: The Impact of Arousal and Valence on Event-Related Potentials.” *International Journal of Psychophysiology* 78, no. 3: 299–307.

- Belin, P., S. Fillion-Bilodeau, and F. Gosselin. 2008. "The Montreal Affective Voices: A Validated Set of Nonverbal Affect Bursts for Research on Auditory Affective Processing." *Behavior Research Methods* 40, no. 2: 531–539.
- Botch, T. L., A. R. Vaidya, and D. Badre. 2024. "Neural Representations of Concreteness and Abstractness Are Organized Into Manifolds." *Journal of Neuroscience* 44, no. 27: e1623–e1677. <https://doi.org/10.1523/JNEUROSCI.1677-23.2024>.
- Brück, C., B. Kreifelts, and D. Wildgruber. 2011. "Emotional Voices in Context: A Neurobiological Model of Multimodal Affective Information Processing." *Physics of Life Reviews* 8, no. 4: 383–403.
- Campanella, S. 2023. "The Potential Utility of Evoked Potentials in the Treatment of Mental Illnesses." *Psychoradiology* 3: kkad024.
- Campbell, R. 2008. "The Processing of Audiovisual Speech: Empirical and Neural Bases." *Philosophical Transactions of the Royal Society, B: Biological Sciences* 363, no. 1493: 1001–1010.
- Chang, C. C., and C. J. Lin. 2011. "LIBSVM: A Library for Support Vector Machines." *ACM Transactions on Intelligent Systems and Technology* 2, no. 3: 1–27.
- Chun, J. W., H. J. Park, I. H. Park, and J. J. Kim. 2012. "Common and Differential Brain Responses in Men and Women to Nonverbal Emotional Vocalizations by the Same and Opposite Sex." *Neuroscience Letters* 515, no. 2: 157–161.
- Cichy, R. M., D. Pantazis, and A. Oliva. 2014. "Resolving Human Object Recognition in Space and Time." *Nature Neuroscience* 17, no. 3: 455–462.
- Cichy, R. M., and D. Pantazis. 2017. "Multivariate Pattern Analysis of MEG and EEG: A Comparison of Representational Structure in Time and Space." *Neuroimage* 158: 441–454.
- Citron, F. M. 2012. "Neural Correlates of Written Emotion Word Processing: A Review of Recent Electrophysiological and Hemodynamic Neuroimaging Studies." *Brain and Language* 122, no. 3: 211–226.
- Citron, F. M., B. S. Weekes, and E. C. Ferstl. 2013. "Effects of Valence and Arousal on Written Word Recognition: Time Course and ERP Correlates." *Neuroscience Letters* 533: 90–95.
- Delaney-Busch, N., G. Wilkie, and G. Kuperberg. 2016. "Vivid: How Valence and Arousal Influence Word Processing Under Different Task Demands." *Cognitive, Affective, & Behavioral Neuroscience* 16: 415–432.
- Delorme, A., and S. Makeig. 2004. "EEGLAB: An Open Source Toolbox for Analysis of Single-Trial EEG Dynamics Including Independent Component Analysis." *Journal of Neuroscience Methods* 134, no. 1: 9–21.
- Doyle, C. M., and K. A. Lindquist. 2018. "When a Word Is Worth a Thousand Pictures: Language Shapes Perceptual Memory for Emotion." *Journal of Experimental Psychology: General* 147, no. 1: 62–73.
- Driver, J., and T. Noesselt. 2008. "Multisensory Interplay Reveals Crossmodal Influences on 'Sensory-Specific' Brain Regions, Neural Responses, and Judgments." *Neuron* 57, no. 1: 11–23.
- Ethofer, T., D. Van De Ville, K. Scherer, and P. Vuilleumier. 2009. "Decoding of Emotional Information in Voice-Sensitive Cortices." *Current Biology* 19, no. 12: 1028–1033.
- Ethofer, T., J. Bretscher, M. Gschwind, B. Kreifelts, D. Wildgruber, and P. Vuilleumier. 2012. "Emotional Voice Areas: Anatomic Location, Functional Properties, and Structural Connections Revealed by Combined fMRI/DTI." *Cerebral Cortex* 22, no. 1: 191–200. <https://doi.org/10.1093/cercor/bhr113>.
- Frühholz, S., and D. Grandjean. 2013. "Processing of Emotional Vocalizations in Bilateral Inferior Frontal Cortex." *Neuroscience & Biobehavioral Reviews* 37, no. 10: 2847–2855.
- Gao, C., D. H. Wedell, and S. V. Shinkareva. 2022. "Crossmodal Negativity Bias in Semantic Processing." *Emotion* 22, no. 6: 1270.
- Gerdes, A., M. J. Wieser, F. Bublatzky, A. Kusay, M. M. Plichta, and G. W. Alpers. 2013. "Emotional Sounds Modulate Early Neural Processing of Emotional Pictures." *Frontiers in Psychology* 4: 741.
- Gerdes, A. B., M. J. Wieser, and G. W. Alpers. 2014. "Emotional Pictures and Sounds: A Review of Multimodal Interactions of Emotion Cues in Multiple Domains." *Frontiers in Psychology* 5: 1351.
- Gianotti, L. R., P. L. Faber, M. Schuler, R. D. Pascual-Marqui, K. Kochi, and D. Lehmann. 2008. "First Valence, Then Arousal: The Temporal Dynamics of Brain Electric Activity Evoked by Emotional Stimuli." *Brain Topography* 20: 143–156.
- Giordano, B. L., S. McAdams, R. J. Zatorre, N. Kriegeskorte, and P. Belin. 2013. "Abstract Encoding of Auditory Objects in Cortical Activity Patterns." *Cerebral Cortex* 23, no. 9: 2025–2037.
- Glaser, W. R., and M. O. Glaser. 1989. "Context Effects in Stroop-Like Word and Picture Processing." *Journal of Experimental Psychology: General* 118, no. 1: 13–42.
- Grootswagers, T., S. G. Wardle, and T. A. Carlson. 2017. "Decoding Dynamic Brain Patterns From Evoked Responses: A Tutorial on Multivariate Pattern Analysis Applied to Time Series Neuroimaging Data." *Journal of Cognitive Neuroscience* 29, no. 4: 677–697.
- Gu, B., B. Liu, D. Beltrán, and M. de Vega. 2023. "ERP Evidence for Emotion-Specific Congruency Effects Between Sentences and New Words With Disgust and Sadness Connotations." *Frontiers in Psychology* 14: 1154442. <https://doi.org/10.3389/fpsyg.2023.1154442>.
- Herbert, C., M. Junghofer, and J. Kissler. 2008. "Event Related Potentials to Emotional Adjectives During Reading." *Psychophysiology* 45, no. 3: 487–498.
- Hofmann, M. J., L. Kuchinke, S. Tamm, M. L. Võ, and A. M. Jacobs. 2009. "Affective Processing Within 1/10th of a Second: High Arousal Is Necessary for Early Facilitative Processing of Negative but Not Positive Words." *Cognitive, Affective, & Behavioral Neuroscience* 9: 389–397.
- Huang, Z., C. Feng, and Q. Qu. 2023. "Predicting Coarse-Grained Semantic Features in Language Comprehension: Evidence From ERP Representational Similarity Analysis and Chinese Classifier." *Cerebral Cortex* 33, no. 13: 8312–8320.
- Jin, Y., Y. Ma, M. Li, and X. Zheng. 2023. "The Influence of Word Concreteness on Acquired Positive Emotion Association: An Event-Related Potential Study." *Acta Psychologica* 240: 104052.
- Kanske, P., and S. A. Kotz. 2007. "Concreteness in Emotional Words: ERP Evidence From a Hemifield Study." *Brain Research* 1148: 138–148.
- King, J. R., and S. Dehaene. 2014. "Characterizing the Dynamics of Mental Representations: The Temporal Generalization Method." *Trends in Cognitive Sciences* 18, no. 4: 203–210.
- Kissler, J., C. Herbert, P. Peyk, and M. Junghofer. 2007. "Buzzwords: Early Cortical Responses to Emotional Words During Reading." *Psychological Science* 18, no. 6: 475–480.
- Kissler, J., C. Herbert, I. Winkler, and M. Junghofer. 2009. "Emotion and Attention in Visual Word Processing—An ERP Study." *Biological Psychology* 80, no. 1: 75–83.
- Kotz, S. A., and S. Paulmann. 2007. "When Emotional Prosody and Semantics Dance Cheek to Cheek: ERP Evidence." *Brain Research* 1151: 107–118.
- Kousta, S. T., G. Vigliocco, D. P. Vinson, M. Andrews, and E. Del Campo. 2011. "The Representation of Abstract Words: Why Emotion Matters." *Journal of Experimental Psychology: General* 140, no. 1: 14–34.
- Kousta, S. T., D. P. Vinson, and G. Vigliocco. 2009. "Emotion Words, Regardless of Polarity, Have a Processing Advantage Over Neutral Words." *Cognition* 112, no. 3: 473–481.
- Kriegeskorte, N., M. Mur, and P. A. Bandettini. 2008. "Representational Similarity Analysis-Connecting the Branches of Systems Neuroscience." *Frontiers in Systems Neuroscience* 4: 249.
- Kuperman, V., Z. Estes, M. Brysbaert, and A. B. Warriner. 2014. "Emotion and Language: Valence and Arousal Affect Word Recognition." *Journal of Experimental Psychology: General* 143, no. 3: 1065–1081.

- Lakens, D., and A. R. Caldwell. 2021. "Simulation-Based Power Analysis for Factorial Analysis of Variance Designs." *Advances in Methods and Practices in Psychological Science* 4, no. 1: 2515245920951503.
- Lavan, N., P. Rinke, and M. Scharinger. 2024. "The Time Course of Person Perception From Voices in the Brain." *Proceedings of the National Academy of Sciences of the United States of America* 121, no. 26: e2318361121. <https://doi.org/10.1073/pnas.2318361121>.
- LeDoux, J. 2012. "Rethinking the Emotional Brain." *Neuron* 73, no. 4: 653–676.
- Li, Y., M. Zhang, S. Liu, and W. Luo. 2022. "EEG Decoding of Multidimensional Information From Emotional Faces." *Neuroimage* 258: 119374.
- Liang, J., M. Zhang, L. Yang, et al. 2025. "How Linguistic and Nonlinguistic Vocalizations Shape the Perception of Emotional Faces—An Electroencephalography Study." *Journal of Cognitive Neuroscience* 37: 970–987.
- Liu, J., L. Fan, L. Tian, C. Li, and W. Feng. 2023. "The Neural Mechanisms of Explicit and Implicit Processing of Chinese Emotion-Label and Emotion-Laden Words: Evidence From Emotional Categorization and Emotional Stroop Tasks." *Language, Cognition and Neuroscience* 38, no. 10: 1412–1429. <https://doi.org/10.1080/23273798.2022.2093389>.
- Liu, Y., N. Liang, D. Wang, et al. 1990. *Dictionary of Usage Frequency of Modern Chinese Words*. Aerospace Press.
- Luck, S. J. 2014. *An Introduction to the Event-Related Potential Technique*. MIT Press.
- Luo, W., and Z. Qi. 2022. "The Influence of Concreteness on Emotional Nouns Valence Processing: An ERP Study." *Acta Psychologica Sinica* 54, no. 2: 111–121.
- Montefinese, M., A. Visalli, A. Angrilli, and E. Ambrosini. 2025. "Fine-Grained Concreteness Effects on Word Processing and Representation Across Three Tasks: An ERP Study." *Psychophysiology* 62, no. 5: e70074. <https://doi.org/10.1111/psyp.70074>.
- Nichols, T. E., and A. P. Holmes. 2002. "Nonparametric Permutation Tests for Functional Neuroimaging: A Primer With Examples." *Human Brain Mapping* 15, no. 1: 1–25.
- Oosterhof, N. N., A. C. Connolly, and J. V. Haxby. 2016. "CoSMoMVPA: Multi-Modal Multivariate Pattern Analysis of Neuroimaging Data in Matlab/GNU Octave." *Frontiers in Neuroinformatics* 10: 27.
- Paivio, A. 1986. *Mental Representations: A Dual Coding Approach*. Oxford University Press.
- Palazova, M., K. Mantwill, W. Sommer, and A. Schacht. 2011. "Are Effects of Emotion in Single Words Non-Lexical? Evidence From Event-Related Brain Potentials." *Neuropsychologia* 49, no. 9: 2766–2775.
- Palazova, M., W. Sommer, and A. Schacht. 2013. "Interplay of Emotional Valence and Concreteness in Word Processing: An Event-Related Potential Study With Verbs." *Brain and Language* 125, no. 3: 264–271.
- Pauligk, S., S. A. Kotz, and P. Kanske. 2019. "Differential Impact of Emotion on Semantic Processing of Abstract and Concrete Words: ERP and fMRI Evidence." *Scientific Reports* 9, no. 1: 14439.
- Pell, M. D., K. Rothermich, P. Liu, S. Paulmann, S. Sethi, and S. Rigoulot. 2015. "Preferential Decoding of Emotion From Human Nonlinguistic Vocalizations Versus Speech Prosody." *Biological Psychology* 111: 14–25.
- Pion-Tonachini, L., K. Kreutz-Delgado, and S. Makeig. 2019. "ICLabel: An Automated Electroencephalographic Independent Component Classifier, Dataset, and Website." *Neuroimage* 198: 181–197.
- Recio, G., M. Conrad, L. B. Hansen, and A. M. Jacobs. 2014. "On Pleasure and Thrill: The Interplay Between Arousal and Valence During Visual Word Recognition." *Brain and Language* 134: 34–43.
- Russell, J. A. 1980. "A Circumplex Model of Affect." *Journal of Personality and Social Psychology* 39, no. 6: 1161–1178.
- Schacht, A., and W. Sommer. 2009. "Time Course and Task Dependence of Emotion Effects in Word Processing." *Cognitive, Affective, & Behavioral Neuroscience* 9, no. 1: 28–43.
- Scherer, L. D., and R. J. Larsen. 2011. "Cross-Modal Evaluative Priming: Emotional Sounds Influence the Processing of Emotion Words." *Emotion* 11, no. 1: 203–208.
- Schindler, S., R. Weidner, and G. R. Fink. 2023. "Dissociating Different Temporal Stages of Emotional Word Processing by Feature-Based Attention." *Scientific Reports* 13: 19299. <https://doi.org/10.1038/s41598-023-43794-4>.
- Steinbeis, N., and S. Koelsch. 2011. "Affective Priming Effects of Musical Sounds on the Processing of Word Meaning." *Journal of Cognitive Neuroscience* 23, no. 3: 604–621.
- Tong, X., C. McBride-Chang, H. Shu, and A. M. Wong. 2009. "Morphological Awareness, Orthographic Knowledge, and Spelling Errors: Keys to Understanding Early Chinese Literacy Acquisition." *Scientific Studies of Reading* 13, no. 5: 426–452.
- Van Hooft, J. C., K. C. Dietz, D. Sharma, and H. Bowman. 2008. "Neural Correlates of Intrusion of Emotion Words in a Modified Stroop Task." *International Journal of Psychophysiology* 67, no. 1: 23–34.
- Vigliocco, G., S. T. Koutsta, P. A. Della Rosa, et al. 2014. "The Neural Representation of Abstract Words: The Role of Emotion." *Cerebral Cortex* 24, no. 7: 1767–1777.
- Vinson, D., M. Ponari, and G. Vigliocco. 2014. "How Does Emotional Content Affect Lexical Processing?" *Cognition & Emotion* 28, no. 4: 737–746.
- Walker, I., and C. Hulme. 1999. "Concrete Words Are Easier to Recall Than Abstract Words: Evidence for a Semantic Contribution to Short-Term Serial Recall." *Journal of Experimental Psychology: Learning, Memory, and Cognition* 25, no. 5: 1256–1271.
- Wang, L., T. Brothers, O. Jensen, and G. R. Kuperberg. 2024. "Dissociating the Pre-Activation of Word Meaning and Form During Sentence Comprehension: Evidence From EEG Representational Similarity Analysis." *Psychonomic Bulletin & Review* 31, no. 2: 862–873.
- Wang, Z. H., and Z. Yao. 2012. "Concreteness Effects of Emotional Noun Words: Evidences From ERP." *Acta Psychologica Sinica* 44, no. 2: 154.
- Weber, V., S. Ruch, N. H. Skieresz, N. Rothen, and T. P. Reber. 2024. "Correlates of Implicit Semantic Processing as Revealed by Representational Similarity Analysis Applied to EEG." *iScience* 27, no. 11: 111149. <https://doi.org/10.1016/j.isci.2024.111149>.
- Yao, B., A. Keitel, G. Bruce, G. G. Scott, P. J. O'Donnell, and S. C. Sereno. 2018. "Differential Emotional Processing in Concrete and Abstract Words." *Journal of Experimental Psychology: Learning, Memory, and Cognition* 44, no. 7: 1064–1074.
- Yao, Z., and Z. Wang. 2014. "Concreteness of Positive Word Contributions to Affective Priming: An ERP Study." *International Journal of Psychophysiology* 93, no. 3: 275–282.
- Yao, Z., D. Yu, L. Wang, X. Zhu, J. Guo, and Z. Wang. 2016. "Effects of Valence and Arousal on Emotional Word Processing Are Modulated by Concreteness: Behavioral and ERP Evidence From a Lexical Decision Task." *International Journal of Psychophysiology* 110: 231–242.
- Zhang, C., Y. Wang, X. Jing, and J. H. Yan. 2023. "Brain Mechanisms of Mental Processing: From Evoked and Spontaneous Brain Activities to Enactive Brain Activity." *Psychoradiology* 3: kka010.
- Zhang, D., W. He, T. Wang, et al. 2014. "Three Stages of Emotional Word Processing: An ERP Study With Rapid Serial Visual Presentation." *Social Cognitive and Affective Neuroscience* 9, no. 12: 1897–1903.

TABLE A1 | Visual word stimuli with Chinese translations, mean valence, arousal, and concreteness ratings, number of strokes, and word frequency.

	Valence	Arousal	Concreteness	Number of strokes	Frequency
Positive words					
Brilliant (辉煌)	7.35	5.65	4	25	1.95
Happiness (幸福)	8	5.85	3.85	21	2.85
Clever (聪明)	7.25	5.3	4.3	23	2.4
Elegant (儒雅)	7.5	5.3	4.4	28	0.7
Handsome (俊美)	7.5	4.8	5.5	18	1.61
Erudite (博学)	7.6	4.85	4.55	20	1.95
Excellence (卓越)	7.35	5.65	4.15	20	1.86
Friendly (友善)	7.25	4.8	3.6	16	1.6
Healthy (健康)	7.5	4.6	4.95	21	1.86
Honest (诚实)	7.6	4.45	4.25	16	1.57
Intelligent (聪颖)	7.65	5.4	3.85	28	3.53
Kind (善良)	7.45	4.7	3.25	19	0.85
Lucky (幸运)	7.65	5.8	3.75	15	2.47
Magnificent (壮丽)	7.65	6.05	4.75	13	2.88
Noble (崇高)	7.85	6.15	3.35	21	0.78
Noble (高尚)	7.4	5.7	3.35	18	1.96
Perfect (完美)	8	5.6	3.75	16	1.9
Perfection (圆满)	7.65	6.2	3.3	23	2.2
Pretty (漂亮)	7.25	5.8	4.9	23	2.05
Righteous (正直)	7.45	4.75	3.5	13	2.11
Selfless (无私)	7.25	4.6	3.45	11	1.18
Sincerity (赤诚)	7.3	5.3	3.65	15	2.38
Sweet (甜蜜)	7.6	5.1	3.9	25	0.85
Victory (胜利)	7.55	6.3	4.55	16	2.27
Wise (睿智)	7.5	5.05	4.3	26	1.69
Negative words					
Barbarous (野蛮)	2.15	5.5	4.15	23	0.7
Cowardly (窝囊)	2.3	4.7	4.4	34	2.03
Degenerate (堕落)	2.2	5.15	3.8	23	1.61
Despair (绝望)	2	6.5	4.25	20	0.7
Dirty (龌龊)	2	6.15	3.7	32	1.15
Failure (失败)	2.15	5.6	4	13	2.17
Fierce (凶恶)	2.05	6.05	3.6	14	1.95
Greedy (贪婪)	2.05	5.45	4.25	19	1.28
Hypocritical (伪善)	1.8	5.65	4	18	2.05
Hypocritical (虚伪)	1.95	5	3.35	17	1.97

(Continues)

TABLE A1 | (Continued)

	Valence	Arousal	Concreteness	Number of strokes	Frequency
Inability (无能)	2.05	5.1	3.75	14	1.99
Miserable (悲惨)	2.05	5.75	3.75	23	1.9
Narrow minded (狭隘)	2.25	5.9	3.4	21	1.7
Poisonous (歹毒)	1.6	6.3	4.3	13	0.78
Desolate (凄凉)	2.25	5.45	3.4	20	2.28
Selfish (自私)	2.3	5.55	3.6	13	3.07
Sorrowful (悲痛)	1.95	6.2	4.2	24	2
Stingy (吝啬)	2.1	5.2	4.3	18	1.58
Thrilling (惊悚)	2.2	7.25	4.45	21	1.96
Treacherous (奸诈)	1.95	5.5	4.2	13	0.78
Vicious (狠毒)	2.3	5.85	3.8	18	1.72
Vulgar (恶俗)	1.8	5.95	4.25	19	1.58
Vulgar (粗俗)	2.25	5	4.55	20	0.78
Vulgar (低俗)	2.25	5	3.9	16	1.8
Obscene (猥琐)	1.85	5.7	4.15	23	1.41
Neutral words					
Actual (现实)	4.7	3.2	4.8	16	1.43
Appropriate (适当)	5.25	2.1	4.95	15	2.11
Uniform (均匀)	5.05	2.2	3.75	11	1.85
Balanced (均衡)	5.25	2.4	4.1	23	2.57
Cream (乳白)	5.3	2.8	5.9	13	1.13
Dispersed (分散)	4.9	2.35	4.75	16	1.75
Fast (快速)	5.35	3.15	4.05	17	3.28
Fixed (固定)	4.9	2.1	4.3	16	2.23
Common (普通)	4.85	1.95	3.15	22	2.78
Global (全球)	5.25	2.95	5.9	17	1.92
Long (长期)	4.9	2.1	3.8	16	1.48
Nearby (附近)	4.95	2.25	4.9	14	3.34
Normal (常态)	4.85	2.3	3.45	19	3.25
Objective (客观)	5.8	3	3.25	15	1.59
Ordinary (平凡)	5.1	3.8	3.05	8	1.08
Overlapping (重合)	5.15	2.65	5.3	15	2.06
Partial (局部)	4.8	1.95	4.75	17	2.01
Random (随机)	4.9	2.2	3.25	17	2.16
Rapid (迅速)	5.65	3.35	4.1	16	1.34
Recent (近期)	5.05	2.1	3.25	19	2.32
Scattered (零散)	4.6	2.5	4.6	25	1.56
Symmetrical (对称)	5.35	2.85	6.1	15	1.23
Temporary (临时)	4.55	2.65	4.35	16	1.52
Thorough (透彻)	5.65	3.95	4.2	17	2.17
Usual (平常)	4.95	2.45	2.8	16	1.49

TABLE A2 | Linguistic vocalization stimuli with Chinese translations and mean valence, arousal, and concreteness ratings.

	Valence	Arousal	Concreteness
Positive words			
Brave	7.4	6.3	4.2
Confident	7.55	5.25	4.1
Delightful	7.4	6.05	3.6
Glad	7.45	6.45	4.2
Happy	7.4	6.3	4.35
Loyal	7.45	5.3	3.75
Optimistic	7.25	5.65	4.05
Outstanding	7.65	5.5	4.05
Pleasant	7.5	6.25	4.35
Successful	7.65	6.15	4.35
Negative words			
Aggressive	2.2	5.6	4.05
Brutal	2.05	6.8	4.35
Cruel	2	6.35	4
Filthy	2.3	6.15	4.6
Defective	1.95	5.35	3.25
Inferior	1.55	6.05	4.4
Scoundrel	2.15	5.85	3.9
Shameless	2.05	5.6	3.9
Terrifying	2.25	7.15	4.85
Ugly	1.95	5.9	3.95
Neutral words			
Abstract	4.85	1.85	2.3
Daily	4.95	1.95	3
Dry	3.9	3.4	5.55
Flat	5.25	2.4	3.85
General	4.6	2.15	2.75
Local	4.9	1.7	5.8
Neutral	5	2.05	3.45
Plain	5	2.6	3.4
Transparent	5.15	2.6	5.95
Unusual	4.85	1.65	3.15

Regarding Stimulus Combinations Under Incongruent Conditions

In the incongruent condition, the priming sound used two other emotional sounds but the same sound type (e.g., linguistic sounds corresponding to linguistic sounds, details in the [Appendix](#)), sounds as incongruent stimuli. For example, in the EEG study, the negative word

nonlinguistic sound incongruence condition consisted of 60 trials, and each negative word (25 pieces) was repeated 2 times, with the remaining 10 trials randomly selected from the 25 negative words. Subsequently, the primed sounds consisted of nonlinguistic positive (10 pieces) and neutral (10 pieces) sounds, repeated three times. Finally, their direct combination was completely randomized. The stimulus combinations for all conditions are shown in the table below.

Condition	Word stimulus	Sound stimulus
Negative-nonlinguistic-congruent	Negative word	Negative nonlinguistic sound
Negative-nonlinguistic-incongruent	Negative word	Positive & neutral nonlinguistic sound
Negative-linguistic-congruent	Negative word	Negative linguistic sound
Negative-linguistic-incongruent	Negative word	Positive & neutral linguistic sound
Positive-nonlinguistic-congruent	Positive word	Positive nonlinguistic sound
Positive-nonlinguistic-incongruent	Positive word	Negative & neutral nonlinguistic sound
Positive-linguistic-congruent	Positive word	Positive linguistic sound
Positive-linguistic-incongruent	Positive word	Negative & neutral linguistic sound
Neutral-nonlinguistic-congruent	Neutral word	Neutral nonlinguistic sound
Neutral-nonlinguistic-incongruent	Neutral word	Positive & negative nonlinguistic sound
Neutral-linguistic-congruent	Neutral word	Neutral linguistic sound
Neutral-linguistic-incongruent	Neutral word	Positive & negative linguistic sounds

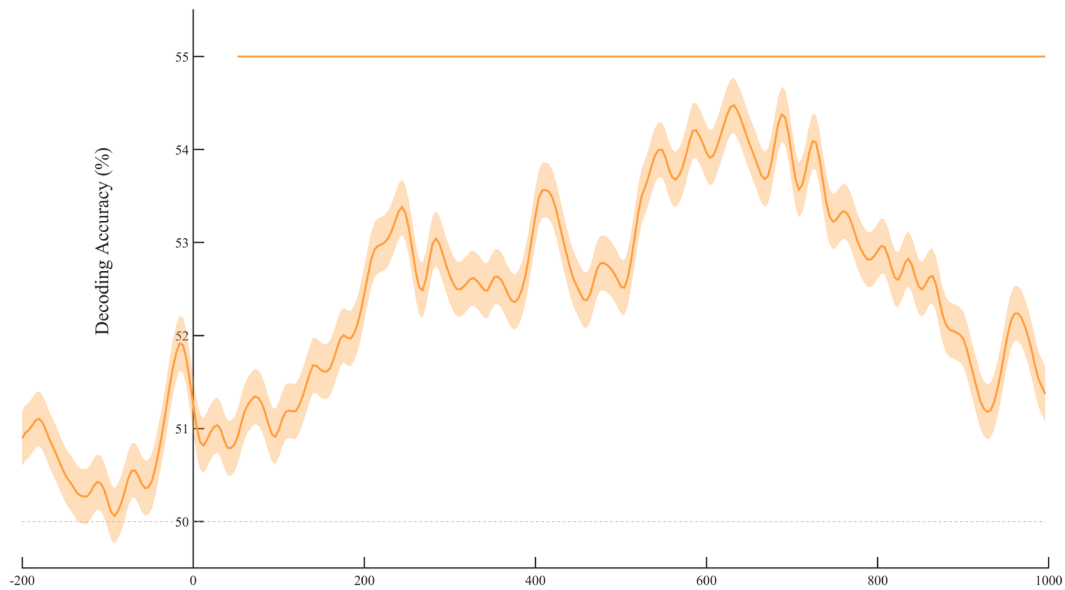


FIGURE A1 | Decoding accuracy. Time course of image decoding, where 0 indicates image onset. The lines above the horizontal axis indicate significant time points using the cluster-corrected sign reciprocity test (cluster definition threshold $p < 0.05$, cluster-corrected significance level $p < 0.05$).